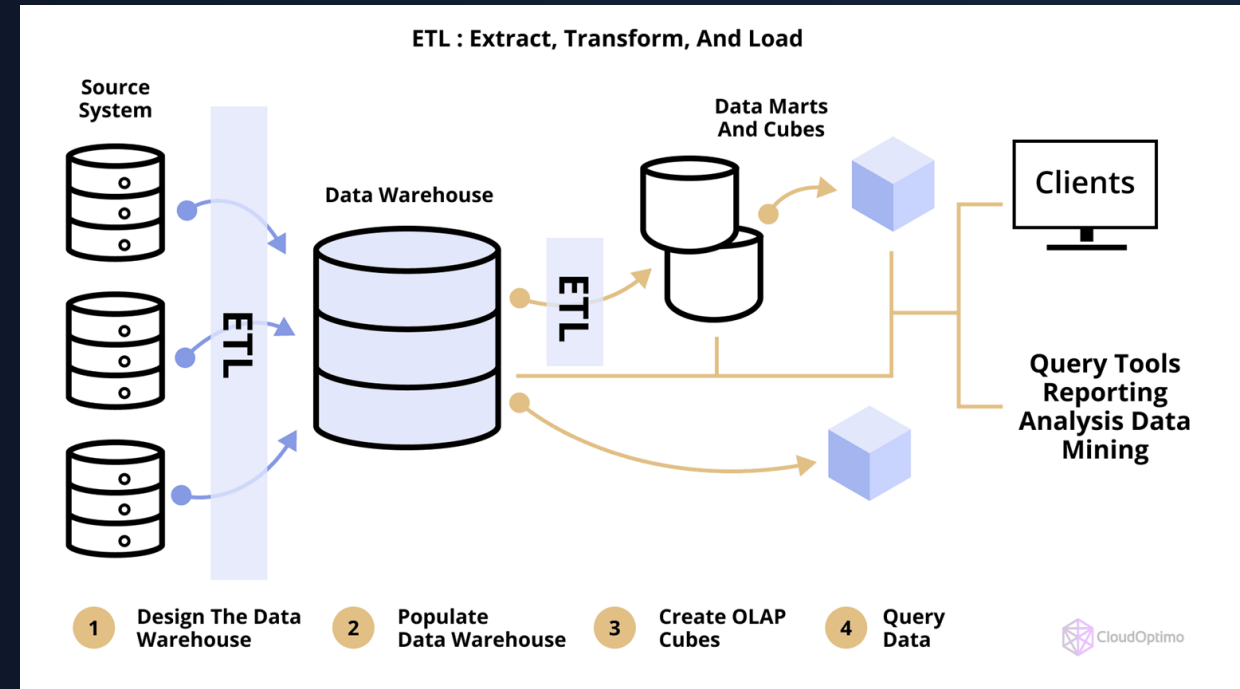


Platform Architecture & Engineering Strategy

- Production-Minded Setup Deployed **PostgreSQL** via Docker, managed entirely by SQLAlchemy and Alembic for robust schema migrations.
- Data Flow & Ingestion ETL securely processes raw telemetry (JSONL) and HR data (CSV) into normalized schemas (employees, events, sessions).
- API & Visualization Layer Data is served via a strict **FastAPI** (/api/v1) contract, directly driving a modular **Streamlit** dashboard.
- Candidate-Led Development Architecture decisions were internally led. LLMs were utilized strictly as accelerators, with manual validation of all logic and API contracts.



Telemetry Insights & Business Value



Pervasive Total Adoption Telemetry reveals 100 active users out of 100 total employees, demonstrating complete integration into daily workflows.



High Event Engagement The workforce generated distinct, measurable sessions resulting in massive telemetry volume, highlighting platform dependency.



Transparent Cost Efficiency Total calculated cost establishes a clear baseline for ROI, proving the scale of operations is manageable.

100/100

ACTIVE USERS

5,000

TOTAL SESSIONS

454,428

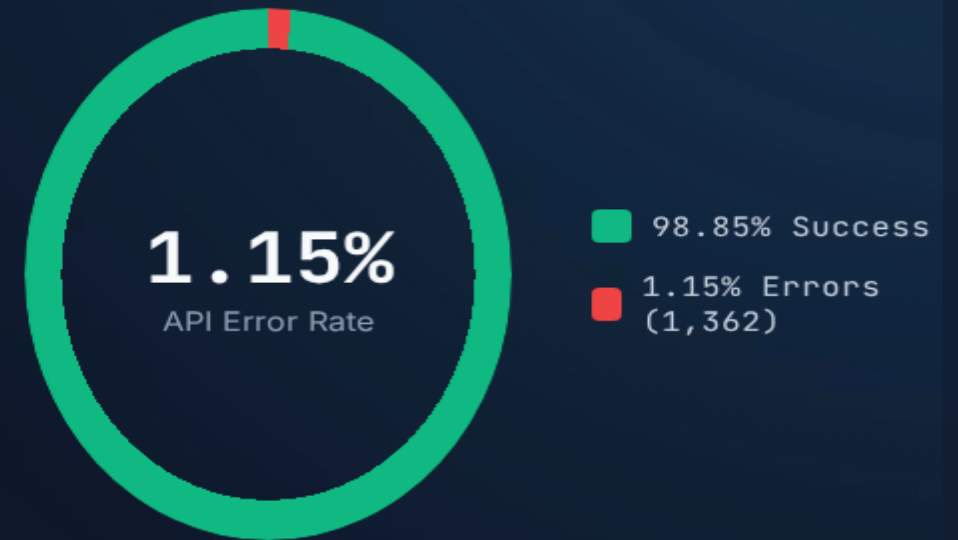
TELEMETRY EVENTS

\$6,001.43

TOTAL COST (USD)

Reliability Analysis & Error Tracking

- 🛡️ **Exceptional System Health** Across nearly half a million events, the platform maintains a near-perfect success rate, proving backend resilience.
- 🔍 **Rigorous Data Validation** Built-in smoke checks ensure ingestion pipelines fail fast on malformed telemetry, protecting the integrity of downstream analytics.
- 🎯 **Actionable Bottleneck Isolation** By isolating the exact 1,362 API errors, engineering can target specific tool usage failures rather than guessing system-wide issues.



Predictive Analytics & Bonus Capabilities



Cost Forecasting Accuracy Implemented linear trend modeling plus weekday seasonality. The holdout backtest yielded a highly accurate MAE of 12.0733 and MAPE of 11.2814.



Automated Anomaly Detection Residual z-score analysis successfully isolated 3 distinct spending anomalies, providing immediate targets for business auditing.



API Extensibility (Bonus) Both standard analytics and predictive models are fully exposed via secure FastAPI endpoints, ready for external consumption.

